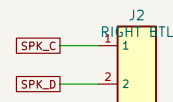
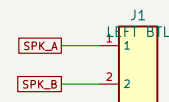
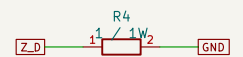
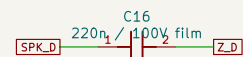
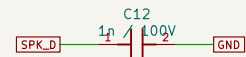
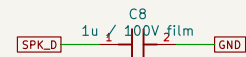
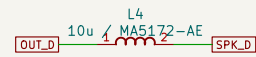
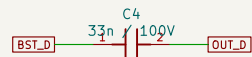
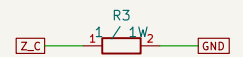
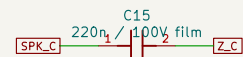
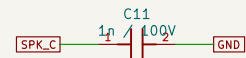
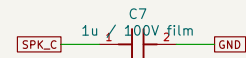
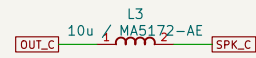
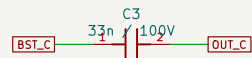
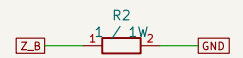
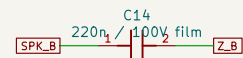
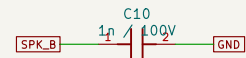
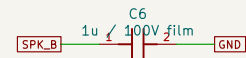
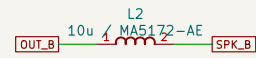
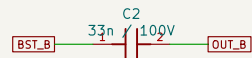
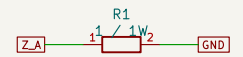
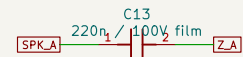
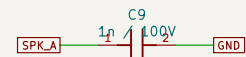
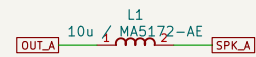
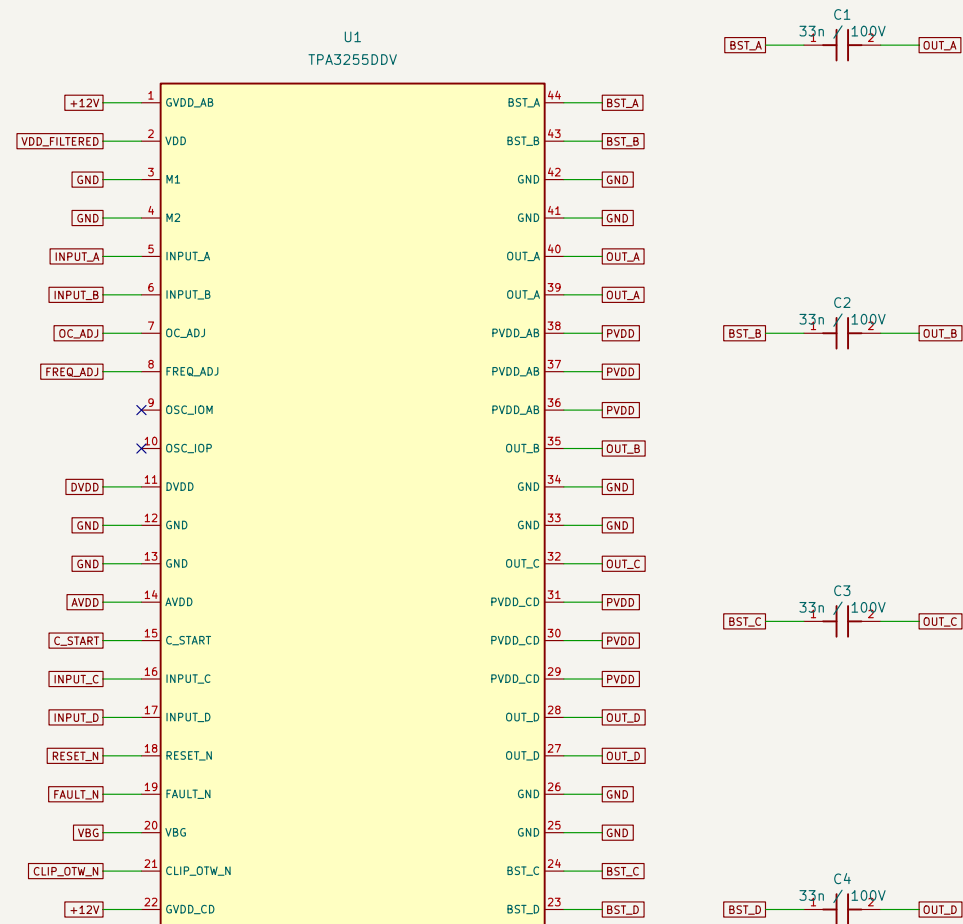


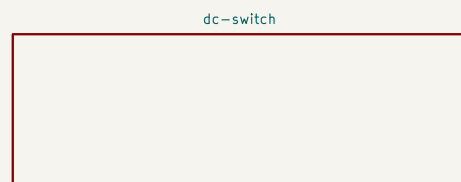
POWER STAGE / STEREO BTL / BENCH PROTOTYPE

Global net names connect all eight sheets. Same labels are electrically connected.



Speaker terminals float: never join B or D to GND.

PFFB + LC: hardware loop stability still needs testing.



File: dc-switch.kicad_sch

Electrical draft – footprints pending / no routing or hardware validation

Sheet: /
File: tpa3255-v03.kicad_sch

Title: TPA3255 V0.3 / tpa3255-v03

Size: A3

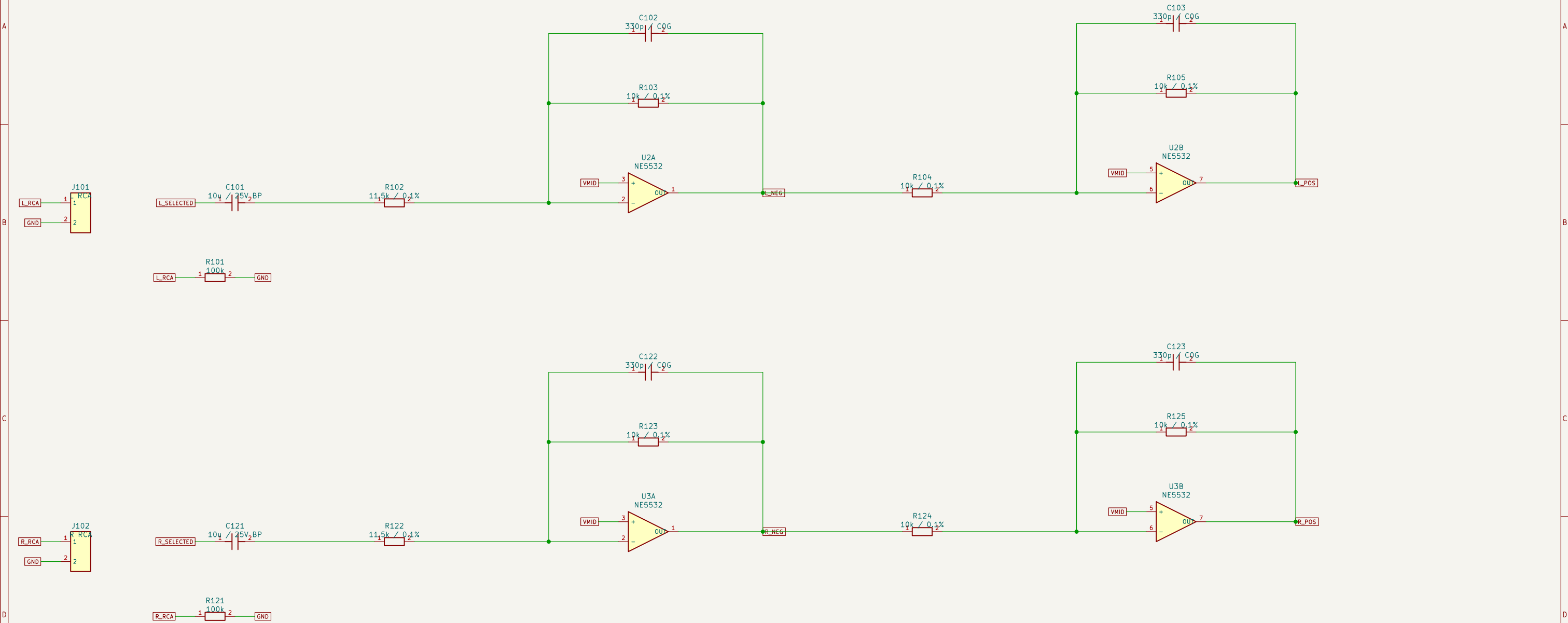
Date: 2026-09-11

Rev: 0.3 REVIEW

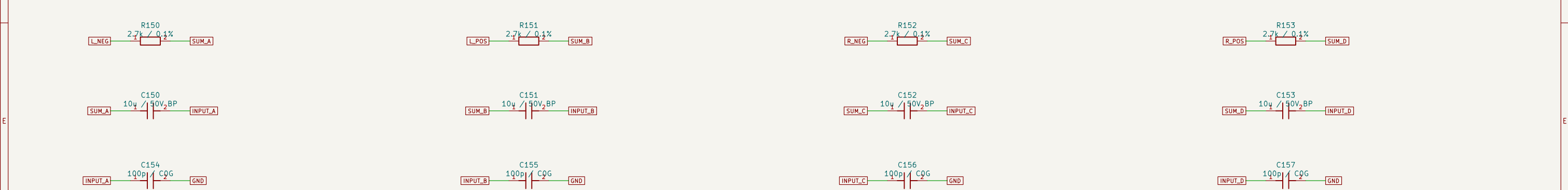
id: 1/8

RCA / XLR SELECTED SIGNAL TO DIFFERENTIAL DRIVER

NE5532 on 12V, VMID approx. 6V. Signal capacitors are bipolar/film.



PFFB summing resistors BEFORE AC coupling; Z10 RCA 2.5Vrms target



Electrical draft – footprints pending / no routing or hardware validation

Sheet: /input/
File: input.kicad_sch

Title: TPA3255 V0.3 / input

Size: A3 Date: 2026-09-11

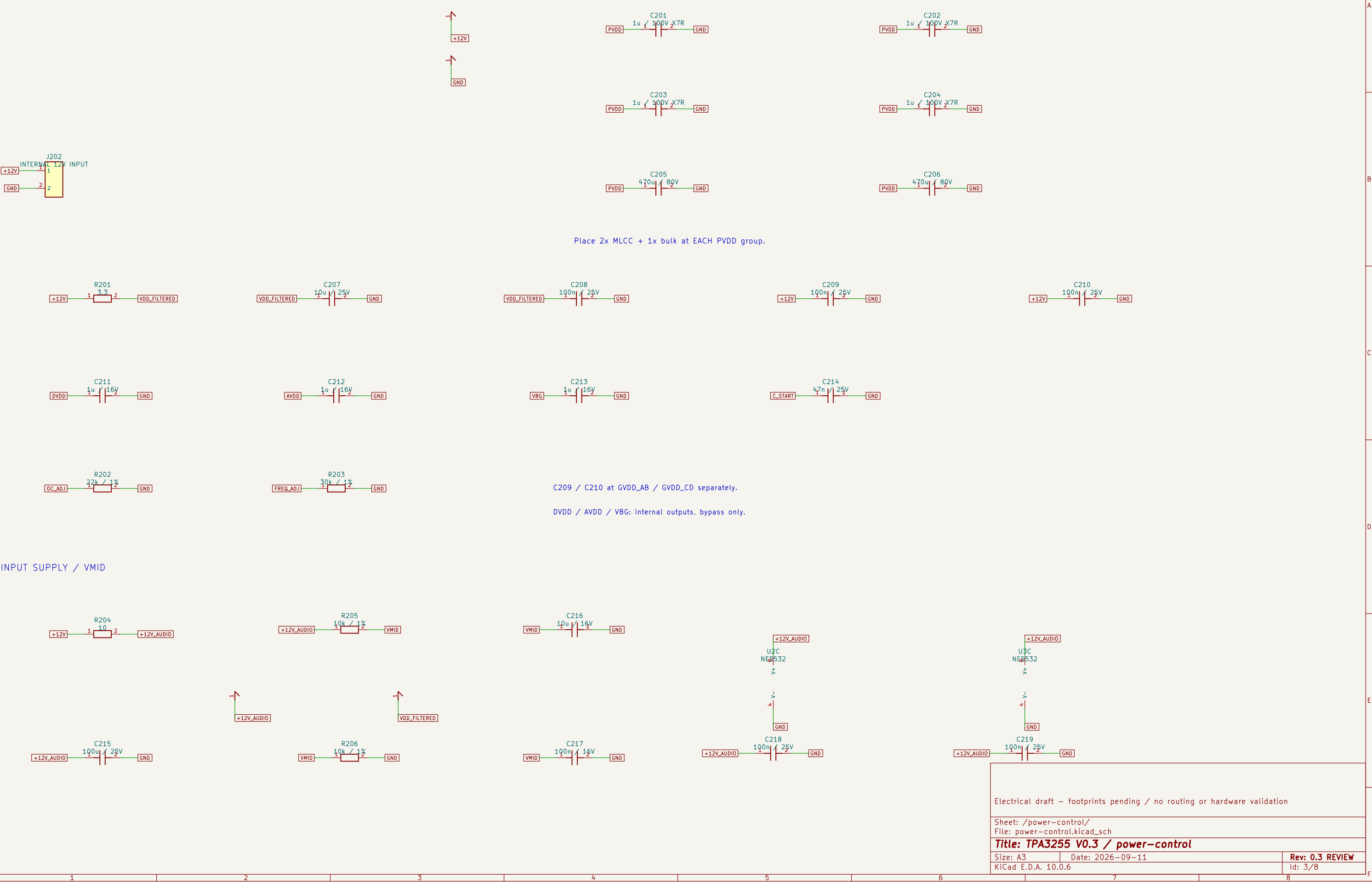
KiCad E.D.A. 10.0.6

Rev: 0.3 REVIEW

Id: 2/8

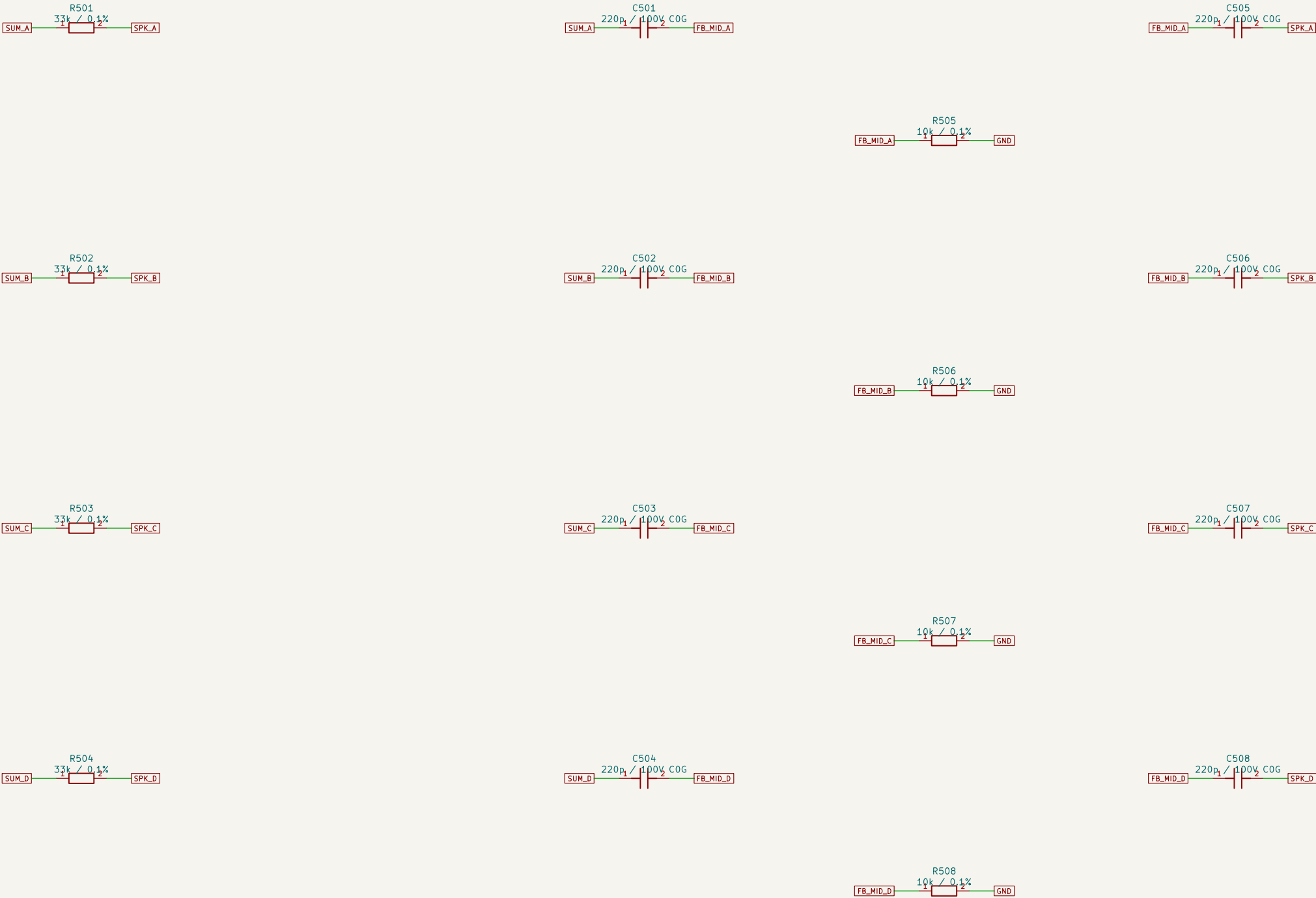
12V STANDBY / LOCAL DECOUPLING

PVDD comes from dc-switch sheet. 12V standby remains on; no mains wiring on this PCB.



POST-FILTER FEEDBACK / TI SLAA788A TPA3255 STARTING VALUES

Same-arm negative feedback: SPK_A to SUM_A, etc. Summing node is BEFORE the input coupling capacitor.



PFFB, input bandwidth and LC are one loop: reduced model is NOT hardware stability proof.

Fit all four feedback branches; do not substitute R/C values without a loop review.

Electrical draft – footprints pending / no routing or hardware validation

Sheet: /feedback/ File: feedback.kicad_sch		
Title: TPA3255 V0.3 / feedback		
Size: A3	Date: 2026-09-11	Rev: 0.3 REVIEW
KiCad E.D.A. 10.0.6		Id: 4/8

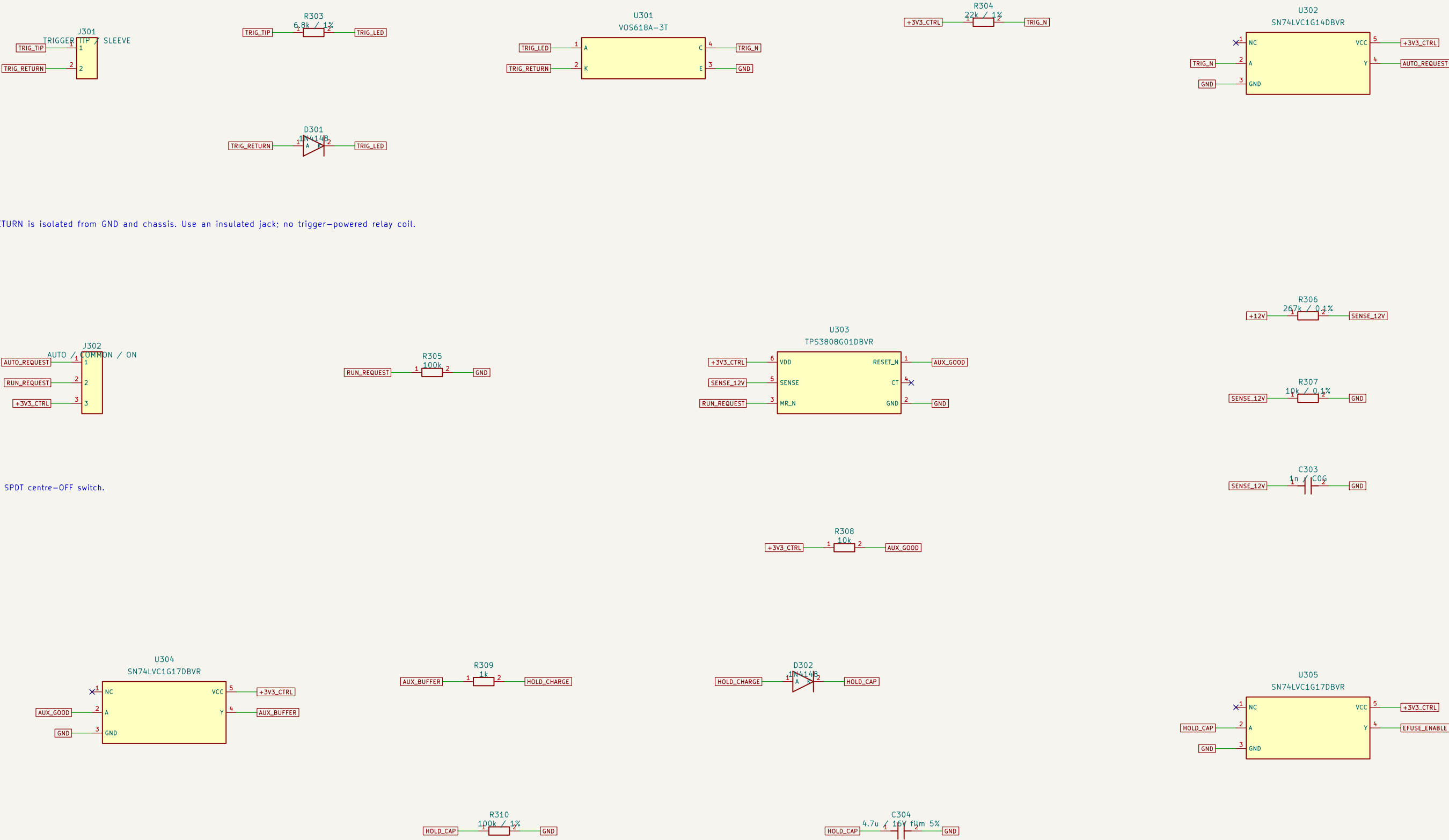
ISOLATED TRIGGER / AUXILIARY MONITOR / POWER-OFF HOLD

Design input: 9–15V DC, nominal 12V. Verify Z10 voltage, polarity and drive capability before connecting.

TRIG_RETURN is isolated from GND and chassis. Use an insulated jack; no trigger-powered relay coil.

External SPDT centre-OFF switch.

AUX_GOOD falls first to mute. HOLD_CAP keeps main DC on briefly; verify timing on hardware.



Electrical draft – footprints pending / no routing or hardware validation

Sheet: /trigger/
File: trigger.kicad_sch

Title: TPA3255 V0.3 / trigger

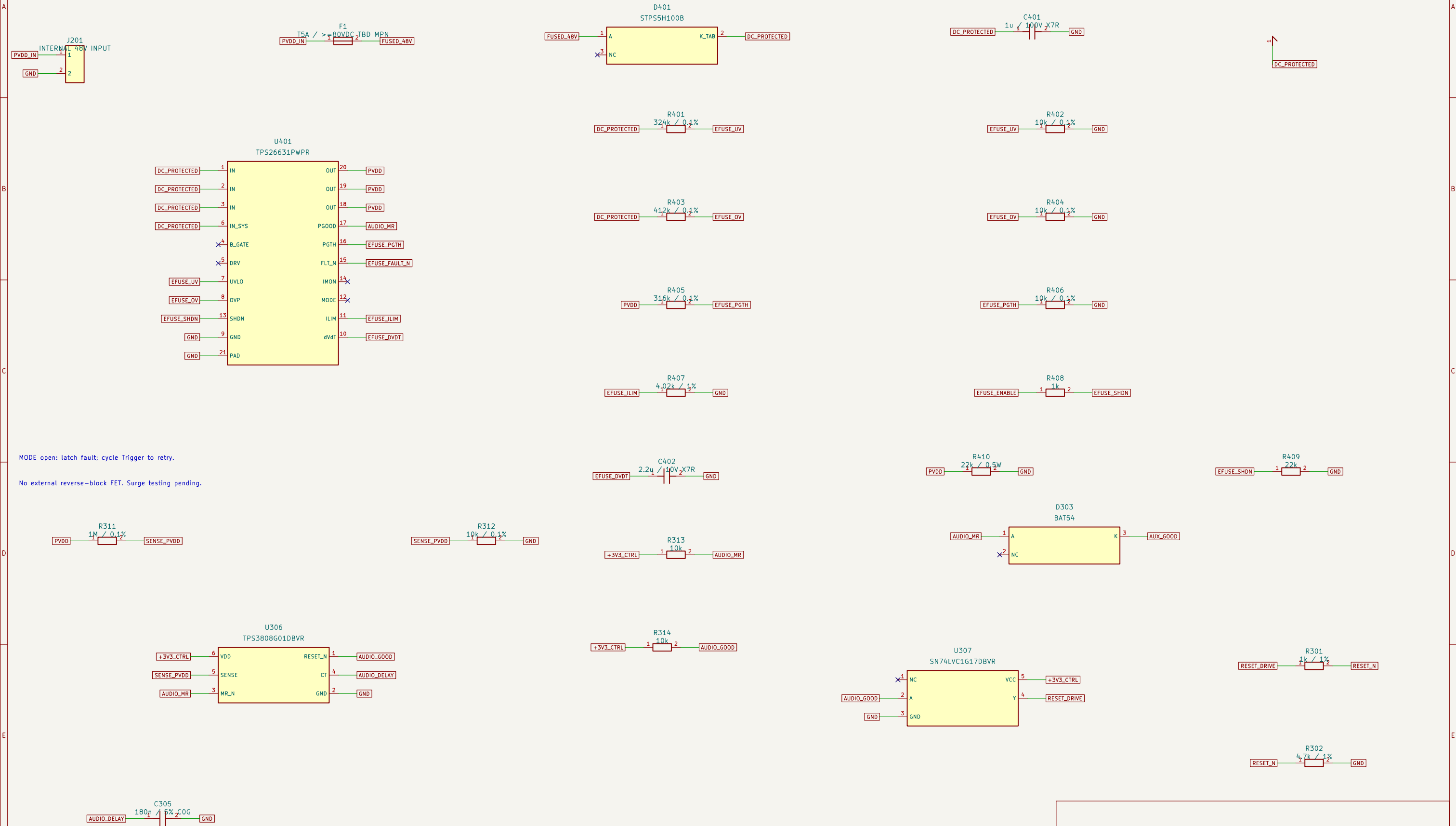
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-11

Rev: 0.3 REVIEW
Id: 5/8

SWITCHED 48V DC / SLEW LIMIT / POWER-GOOD / AUDIO RESET

Internal AC/DC modules share one mains inlet. Trigger-off disconnects PVDD; 12V standby remains on.



MODE open: latch fault; cycle Trigger to retry.

No external reverse-block FET. Surge testing pending.

PVDD >40.9V, then ~1.03s delay.

Electrical draft – footprints pending / no routing or hardware validation

Sheet: /dc-switch/
File: dc-switch.kicad_sch

Title: TPA3255 V0.3 / dc-switch

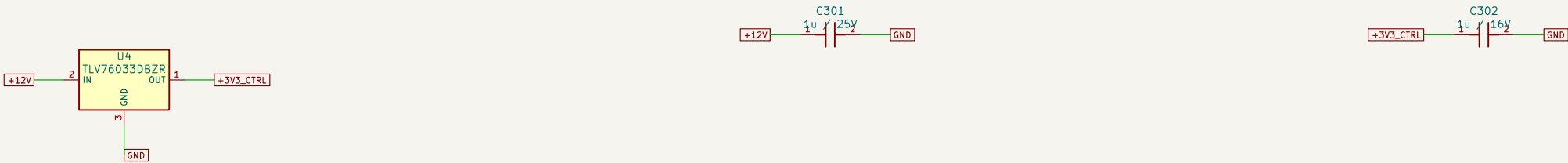
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Rev: 0.3 REVIEW

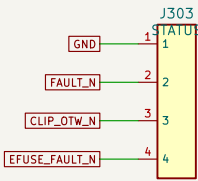
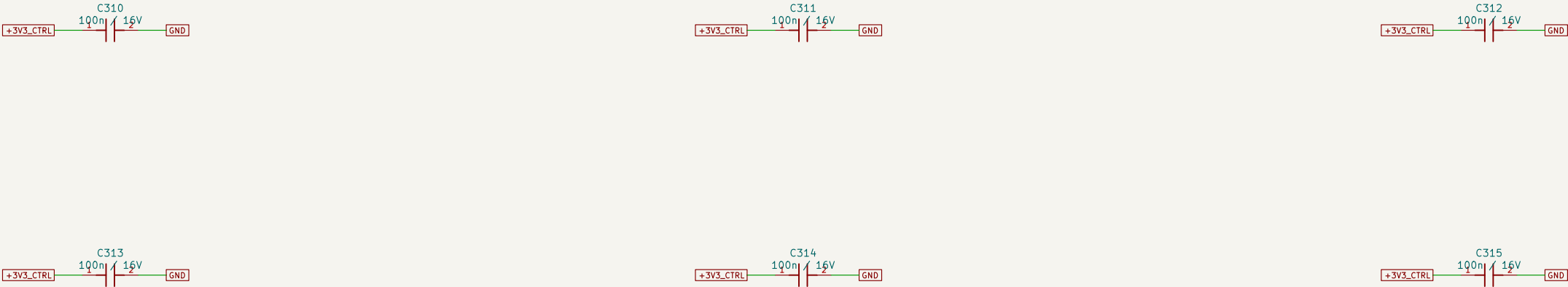
KiCad E.D.A. 10.0.6

Id: 6/8

STANDBY CONTROL SUPPLY / LOCAL DECOUPLING / STATUS



C310–C315: one at each U302–U307 VCC/VDD pin on PCB.



High-impedance monitoring only; no external 5V pullups.

TPA fault protection remains internal; no automatic fault retry loop.

Trigger OFF enters standby, not AC mains isolation.

Electrical draft – footprints pending / no routing or hardware validation

Sheet: /control-supply/
File: control-supply.kicad_sch

Title: TPA3255 V0.3 / control-supply

Size: A3 Date: 2026-09-11

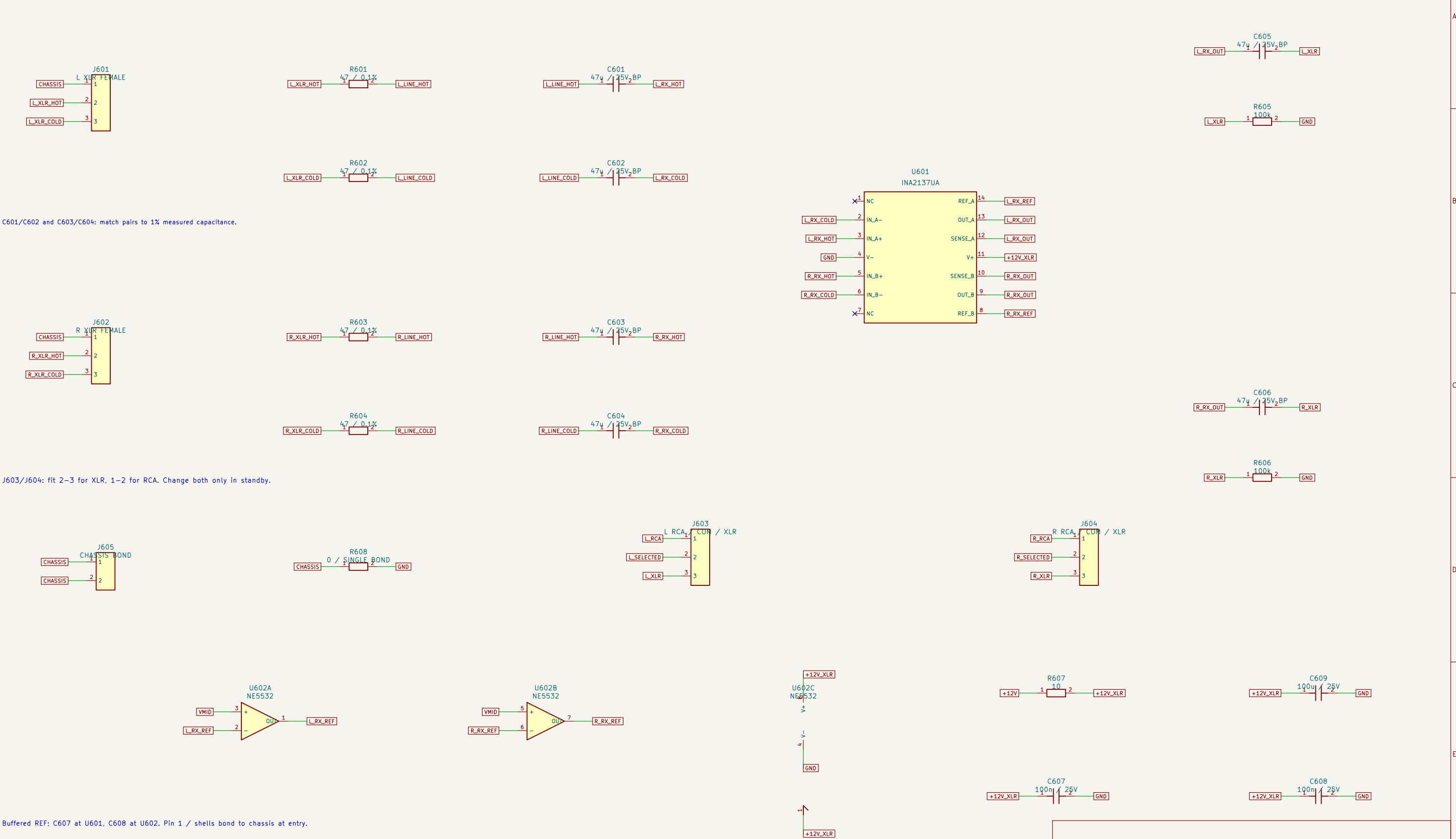
KiCad E.D.A. 10.0.6

Rev: 0.3 REVIEW

Id: 7/8

TRUE BALANCED XLR / INA2137 GAIN 0.5 / RCA SELECTION

XLR female: 1 shield, 2 hot, 3 cold. Both signal legs are AC coupled; do not ground pin 3.



Buffered REF; C607 at U601, C608 at U602. Pin 1 / shells bond to chassis at entry.

Internal balanced-to-single-ended receiver; this is not a fully differential end-to-end amplifier.

Electrical draft – footprints pending / no routing or hardware validation